

Math 4441: Probability and Statistics

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Chapter 2: Random Variables (Part 3)

Operations on Random Variables

Lectures 13 – 14

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Topics Covered

1. Expected value
2. Derived Random Variables
3. Variance
4. n -th Moments
5. n -th Central Moments
6. Covariance
7. Correlations

Expected Values

Example: Suppose you play a game, where in each play you lose \$1 with probability 0.60 and you win \$1, \$2 and \$3 with probabilities 0.30, 0.08 and 0.02, respectively. Find your net gain in each play of the game.

- If you play only a few times, your gain might depend on your luck
- If you play a large number of times – your luck no longer determines the total gain – rather it will be multiple of average gain

Say, you play the game n times, approximately in $0.6 \times n$ games you will loose \$1 per game and will win \$1, \$2 and \$3 in $0.3 \times n$, $0.08 \times n$ and $0.02 \times n$ games, respectively.

Your total gain is

$$(0.6) \times n \cdot (-1) + (0.3) \times n \cdot (1) + (0.08) \times n \cdot (2) + (0.02) \times n \cdot (3) = (-0.08) \times n.$$

Average loss of \$0.08 per game

The more you play, the less luck interferes and your loss comes to \$0.08 per game

- Let X denote the gain in one play
- Then -0.08 is the expected value of X
- $E[X], \mu_X$, or μ is the expected value of X

x	-1	1	2	3	
$P_X(x)$	0.60	0.30	0.08	0.02	1.00
$x \cdot P_X(x)$	-0.60	0.30	0.16	0.06	-0.08

$$P_X(x) = \begin{cases} 0.60, & x = -1 \\ 0.30, & x = 1 \\ 0.08, & x = 2 \\ 0.02, & x = 3 \\ 0, & o/w \end{cases}$$

$$-1 \times 0.6n + 1 \times 0.3n + 2 \times 0.08n + 3 \times 0.02n = -0.08n$$

$$\text{Or, } -1 \times 0.6 + 1 \times 0.3 + 2 \times 0.08 + 3 \times 0.02 = -0.08$$

$$\text{Or, } -1 \times P_X(-1) + 1 \times P_X(1) + 2 \times P_X(2) + 3 \times P_X(3) = -0.08$$

$$E[X] = \sum_{x \in S_X} x \cdot P_X(x)$$

Definition (Expected Value): The *expected value* of a discrete random variable X with the set of possible values S_X and probability mass function $P_X(x)$ is defined by

$$E[X] = \sum_{x \in S_X} x P_X(x)$$

$E[X]$ exists if this sum converges

The expected value of a random variable X is also called the **mean**, or the **mathematical expectation**, or simply the **expectation** of X . It is denoted by $E[X]$, μ_X , or simply μ .

Example: Three packets are sent from one computer to another. Each packet is successfully sent with a probability of 0.8 irrespective of the others. Each successfully sent packet results in a reward of 1 unit whereas 1 unit penalty (negative reward) is imposed

Let X is the net reward. X is a discrete random variable with values

$$S_X = \{-1, 1, 2, 3\}$$

$$P[X = -1] = 0.008$$

$$P[X = 1] = 0.096$$

$$P[X = 2] = 0.384$$

$$P[X = 3] = 0.512$$

$$E[X] = -1 \times 0.008 + 1 \times 0.096 + 2 \times 0.384 + 3 \times 0.512$$

Example: A college mathematics department sends 8 to 12 professors to the annual meeting of the American Mathematical Society, which lasts five days. The hotel at which the conference is held offers a bargain rate of a dollars per day per person if reservations are made 45 or more days in advance, but charges a cancellation fee of $2a$ dollars per person. The department is not certain how many professors will go. However, from past experience it is known that the probability of the attendance of i professors is $1/5$ for $i = 8, 9, 10, 11$, and 12 . If the regular rate of the hotel is $2a$ dollars per day per person, should the department make any reservations? If so, how many?

$$E(X_8) = (40a)\frac{1}{5} + (50a)\frac{1}{5} + (60a)\frac{1}{5} + (70a)\frac{1}{5} + (80a)\frac{1}{5} = 60a$$

$$E(X_9) = (42a)\frac{1}{5} + (45a)\frac{1}{5} + (55a)\frac{1}{5} + (65a)\frac{1}{5} + (75a)\frac{1}{5} = 56.4a,$$

$$E(X_{10}) = (44a)\frac{1}{5} + (47a)\frac{1}{5} + (50a)\frac{1}{5} + (60a)\frac{1}{5} + (70a)\frac{1}{5} = 54.2a,$$

$$E(X_{11}) = (46a)\frac{1}{5} + (49a)\frac{1}{5} + (52a)\frac{1}{5} + (55a)\frac{1}{5} + (65a)\frac{1}{5} = 53.4a,$$

$$E(X_{12}) = (48a)\frac{1}{5} + (51a)\frac{1}{5} + (54a)\frac{1}{5} + (57a)\frac{1}{5} + (60a)\frac{1}{5} = 54a.$$

X_{11} has the smallest expected value. Therefore, making 11 reservation is the most reasonable policy

Derived Random Variable:

If x is a sample value of a random variable X , and y is a sample value of a random variable Y . If we have a function $Y = g(X)$ and we get the values of Y from the values of X . Because we obtain Y from another random variable, we refer to Y as a **derived random variable**.

Definition 2.4 (Derived Random Variable). If X and Y are two random variables, and each sample value y of Y is a mathematical function $g(x)$ of X , then the relationship of these two random variables is given by

$$Y = g(X),$$

and the random variable Y is called a derived random variable.

What will be the PMF of Y ?

For a discrete random variable Y , if there exists another random variable X , where $Y = g(X)$, then the PMF of X is

$$P_Y(y) = \sum_{x:g(x)=y} P_X(x)$$

Expected Value of a derived random variable

$$E[X + Y] = ?$$

$$E[aX + b] = ?$$

n -th Moment of a Random Variable

Variance: Variance measures the average magnitude of the fluctuations of a random variable from its expectations.

$$E[X - E[X]] = E[X - \mu] = E[X] - \mu = E[X] - E[X] = 0.$$

Hence, $E[X - E[X]]$ is not an appropriate measure for the variance.

if we consider $E(|X - E[X]|)$ instead, the problem of negative and positive deviations canceling each other disappears.

mathematically, $E(|X - E[X]|)$ is difficult to handle

The quantity $E[(X - E[X])^2]$ is used to measure the average amount of fluctuations, and is called the variance of X

The square root of $E[(X - E[X])^2]$ is called the **standard deviation** of X .

Definition 2.7. Let X be a discrete random variable with a set of possible values S_X , probability mass function $P_X(x)$, and $E(X) = \mu$. Then σ_X and $Var[X]$, called the **standard deviation** and the **variance** of X , respectively, and are defined by

$$\sigma_X = \sqrt{E[(X - \mu)^2]},$$

and

$$Var[X] = E[(X - \mu)^2].$$

Note that by this definition, we have

$$Var[X] = E[(X - \mu)^2] = \sum_{x \in S_X} (x - \mu)^2 P_X(x).$$

$$\text{Var}[X] = E[X^2] - (E[X])^2.$$

Proof.

$$\begin{aligned}\text{Var}[X] &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2 \\ &= E[X^2] - (E[X])^2.\end{aligned}$$

Var[X] = 0 if and only if X is a constant with probability 1.

$$\text{Var}[aX + b] = a^2 \text{Var}[X]$$

$$P_X(x) = \begin{cases} 0.60, & x = -1 \\ 0.30, & x = 1 \\ 0.08, & x = 2 \\ 0.02, & x = 3 \\ 0, & o/w \end{cases}$$

$$E[X] = \mu = 0.08$$

$$V[X] = [-1 - (-0.08) \times 0.60 + (1 - 0.08) \times 0.30 + (2 - 0.08) \times 0.08 + (3 - .08) \times 0.02$$

n-th Central Moment

Expected Value and Variance for Continuous Random Variables

$$E[X] = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx$$

$$V[X] = \int_{-\infty}^{+\infty} (x - \mu)^2 \cdot f_X(x) dx$$

Example: The time elapsed, in minutes, between the placement of an order of pizza and its delivery is random with pdf

$$f(x) = \begin{cases} 1/15 & \text{if } 25 < x < 40 \\ 0 & \text{otherwise.} \end{cases}$$

$$E(X) = \int_{25}^{40} x \frac{1}{15} dx = 32.5,$$

$$E(X^2) = \int_{25}^{40} x^2 \frac{1}{15} dx = 1075.$$

$$V[X] = 1075 - (32.5)^2 = 18.75$$

$$\sigma_X = \sqrt{V[X]} = 4.33$$

Sketch the graph of the function

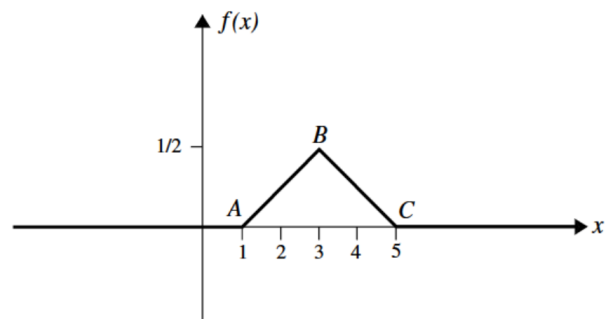
$$f(x) = \begin{cases} \frac{1}{2} - \frac{1}{4}|x - 3| & 1 \leq x \leq 5 \\ 0 & \text{otherwise,} \end{cases}$$

and show that it is the probability density function of a random variable X .

Find F , the distribution function of X , and show that it is continuous.

Find $E[X]$ and $V[X]$.

$$f(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{2} + \frac{1}{4}(x-3) & 1 \leq x < 3 \\ \frac{1}{2} - \frac{1}{4}(x-3) & 3 \leq x < 5 \\ 0 & x \geq 5. \end{cases}$$



For $t < 1$,

$$F(t) = \int_{-\infty}^t 0 \, dx = 0;$$

for $1 \leq t < 3$,

$$F(t) = \int_{-\infty}^t f(x) \, dx = \int_1^t \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx = \frac{1}{8}t^2 - \frac{1}{4}t + \frac{1}{8};$$

for $3 \leq t < 5$,

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) \, dx = \int_1^3 \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx + \int_3^t \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \left(-\frac{1}{8}t^2 + \frac{5}{4}t - \frac{21}{8} \right) = -\frac{1}{8}t^2 + \frac{5}{4}t - \frac{17}{8}; \end{aligned}$$

and for $t \geq 5$,

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) \, dx = \int_1^3 \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx + \int_3^5 \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

Expected Value and Variance from JPMF

Definition 4.3. If the joint PMF of X and Y is $P_{XY}(x, y)$, then $E[X]$ and $E[Y]$ are calculated as

$$\begin{aligned} E[X] &= \sum_x x P_X(x) \\ &= \sum_x x \left[\sum_y P_{XY}(x, y) \right] \\ &= \sum_x \sum_y x P_{XY}(x, y) \end{aligned}$$

Similarly,

$$E[Y] = \sum_y \sum_x y P_{XY}(x, y).$$

Definition 4.4. If the joint PMF of X and Y is $P_{XY}(x, y)$, then $Var[X]$ calculated as

$$\begin{aligned} Var[X] &= \sum_x (x - \mu_x)^2 P_X(x) \\ &= \sum_x (x - \mu_x)^2 \left[\sum_y P_{XY}(x, y) \right] \\ &= \sum_x \sum_y (x - \mu_x)^2 P_{XY}(x, y). \end{aligned}$$

Similarly,

$$Var[Y] = \sum_y \sum_x (y - \mu_y)^2 P_{XY}(x, y)$$

Expected value of a Function of two random variables

For random variables X and Y , the expected value of $W = g(X, Y)$ is

$$\text{Discrete: } E[W] = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) P_{X,Y}(x, y);$$

$$\text{Continuous: } E[W] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) \, dx dy.$$

$$E[a_1 g_1(X, Y) + \cdots + a_n g_n(X, Y)] = a_1 E[g_1(X, Y)] + \cdots + a_n E[g_n(X, Y)]$$

For any two random variables

$$E[X + Y] = E[X] + E[Y]$$

Covariance and correlation

- The variance of a RV X , $V[X] = E[(X - \mu_X)^2]$, represents the average fluctuation of the values of the RV from its expectation μ_X
- It measures the dispersion or spread of the distribution of X about its mean
- For two jointly distributed RVs X and Y , the $V[X]$ and $V[Y]$ determine the individual dispersions or spreads of X and Y independently
 - $V[X]$ measures the dispersion along the x -axis
 - $V[Y]$ measures the dispersion along the y -axis
 - The joint dispersion is missing
- The joint dispersion or spread of X and Y can be measured along the direction $(ax + by)$ by the variance $V[aX + bY]$

$$\begin{aligned}
Var(aX + bY) &= E[(aX + bY)E(aX + bY)]^2 \\
&= E[(aX + bY)aE(X)bE(Y)]^2 \\
&= E\left[a(X - E[X]) + b(Y - E[Y])\right]^2 \\
&= E\left[a^2(X - E[X])^2 + b^2(Y - E[Y])^2 + 2ab(X - E[X])(Y - E[Y])\right]^2 \\
&= a^2Var[X] + b^2Var[Y] + 2abE\left[(X - E[X])(Y - E[Y])\right].
\end{aligned}$$

The joint dispersion or spread of X and Y can be measured in any direction ($ax + by$) if we know

- Variance of X , $V[X]$ Determine the dispersion of X and Y independently
- Variance of Y , $V[Y]$
- $E[(X - \mu_X)(Y - \mu_Y)]$ Should give the measure of the joint dispersion

Definition 4.10. *Covariance.* Let X and Y be random variables defined over the same probability space and such that $E[X] < \infty$ and $E[Y] < \infty$. The covariance between X and Y is defined by:

$$\text{Cov}[X, Y] = E\left[(X - E[X])(Y - E[Y])\right]. \quad (4.39)$$

Note that $\text{Cov}[X, X] = \text{Var}[X]$.

Covariance random variables X and Y is also denoted by C_{XY} .

Let a and b be real numbers; for random variables X and Y ,

$$\text{Var}[aX + bY] = a^2\text{Var}[X] + b^2\text{Var}[Y] + 2ab\text{Cov}[X, Y].$$

In particular, if $a = 1$ and $b = 1$, this gives

$$\text{Var}[X + Y] = a^2\text{Var}[X] + b^2\text{Var}[Y] + 2\text{Cov}[X, Y].$$

Similarly, if $a = 1$ and $b = -1$, it gives

$$\text{Var}[X - Y] = a^2\text{Var}[X] + b^2\text{Var}[Y] - 2\text{Cov}[X, Y].$$

Letting $\mu_X = E[X]$, and $\mu_Y = E[Y]$, an alternative formula for $\text{Cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$ is calculated by the expansion of $E[(X - E[X])(Y - E[Y])]$:

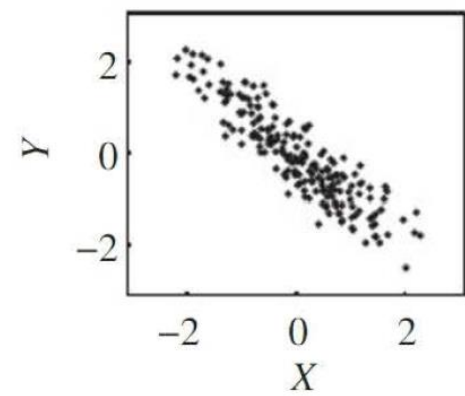
$$\begin{aligned}\text{Cov}[X, Y] &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y] \\ &= E[XY] - \mu_X E[Y] - \mu_Y E[X] + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

For random variables X and Y , the C_{XY} might be

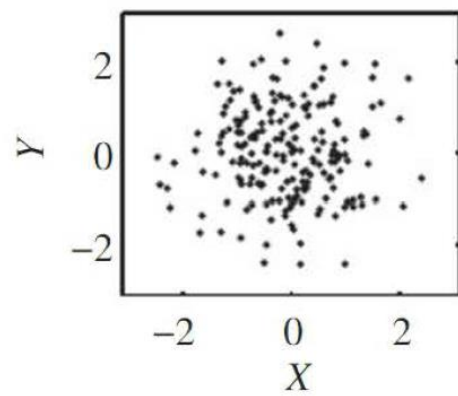
- Positive: X and Y are positively correlated – increase or decrease together
- Negative: X and Y are negatively correlated – X increases while Y decreases, or vice versa
- Zero: X and Y are uncorrelated

Example:

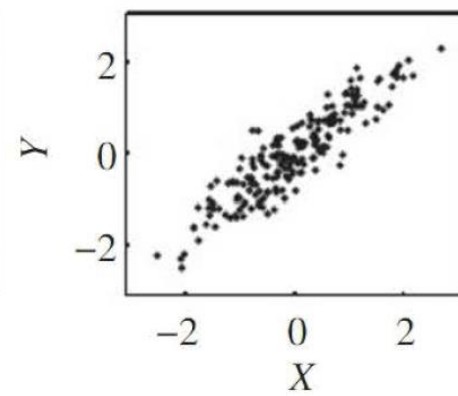
- The Study-time of a student and his CGPA
- The distance of a student from white board and his CGPA



(a) $\rho_{X,Y} = -0.9$



(b) $\rho_{X,Y} = 0$



(c) $\rho_{X,Y} = 0.9$

Example: Suppose we perform an experiment in which we measure X and Y in centimeters (e.g., the height of two students). However, if we change units and measure the height in meters, we will perform the same experiment except we observe $\hat{X} = X/100$ and $\hat{Y} = Y/100$. In this case $\hat{X}$ and $\hat{Y}$ have covariance

$$\begin{aligned}\text{Cov}[\hat{X}\hat{Y}] &= E[(\hat{X} - \mu_{\hat{X}})(\hat{Y} - \mu_{\hat{Y}})] \\ &= \frac{E[(X - \mu_X)(Y - \mu_Y)]}{10000} \\ &= \frac{\text{Cov}[X, Y]}{10000} \text{m}^2.\end{aligned}$$

Changing the unit of measurement from cm^2 to m^2 reduces the covariance by a factor of 10,000.

Correlation Coefficient

A normalized version of C_{XY} , that indicates the relationship of two RVs variables regardless of the measurement units, is called the correlation coefficient and is denoted by ρ_{XY} .

$$\rho_{X,Y} = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X]\text{Var}[Y]}} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}$$

Properties of Covariance:

- If X and Y are independent RVs with $0 < \sigma_X^2 < \infty$ and $0 < \sigma_Y^2 < \infty$, then
$$C_{XY} = \rho_{XY} = 0$$

However, the converse of the above property may not be true.

Example 4.11. Let X be uniformly distributed over $(1, 1)$ and $Y = X^2$. Then

$$\text{Cov}[X, Y] = E[X^3] - E[X]E[X^2] = 0.$$

since $E[X] = 0$ and $E[X^3] = 0$. Thus the perfectly related random variables X and Y are uncorrelated.

- If $X_1, \dots, X_n$ are RVs such that $V[X_i] < \infty$, for $i = 1, 2, \dots, n$

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

- If $X_1, \dots, X_n$ are uncorrelated RVs

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i).$$

- If $X_1, \dots, X_n$ are RVs

$$\text{Var} \left(\sum_{i=1}^n a_i X_i \right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j)$$

In particular, for $a_i = 1, 1 \leq i \leq n$

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Example 4.12. Let X be the lifetime of an electronic system and Y be the lifetime of one of its components. Suppose that the electronic system fails if the component does (but not necessarily vice versa). Furthermore, suppose that the joint probability density function of X and Y (in years) is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{49}e^{-\frac{y}{7}}, & \text{for } 0 \leq x \leq y \leq \infty; \\ 0, & \text{otherwise.} \end{cases}$$

- (1) Determine the expected value of the remaining lifetime of the component when the system dies.
- (2) Find the covariance of X and Y .

(1) The remaining lifetime of the component when the system dies is $Y - X$. So desired quantity is

$$\begin{aligned} E[Y - X] &= \int_0^\infty \int_0^y (y - x) \frac{1}{49} e^{-\frac{y}{7}} dx dy \\ &= \frac{1}{49} \int_0^\infty e^{-\frac{y}{7}} \left(y^2 - \frac{y^2}{2} \right) dy \\ &= \frac{1}{98} \int_0^\infty y^2 e^{-\frac{y}{7}} dy \\ &= 7. \end{aligned}$$

(2) To find $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$, note that

$$\begin{aligned} E[XY] &= \int_0^\infty \int_0^y (xy) \frac{1}{49} e^{-\frac{y}{7}} dx dy \\ &= \frac{1}{49} \int_0^\infty y e^{-\frac{y}{7}} \left(\int_0^y x dx \right) dy \\ &= \frac{1}{98} \int_0^\infty y^3 e^{-\frac{y}{7}} dy \\ &= \frac{14406}{98} \\ &= 147. \end{aligned}$$

$$E[X] = \int_0^\infty \int_0^y x \frac{1}{49} e^{-\frac{y}{7}} dx dy = 7,$$

$$E[Y] = \int_0^\infty \int_0^y y \frac{1}{49} e^{-\frac{y}{7}} dx dy = 14.$$

Therefore, $\text{Cov}[X, Y] = 1477(14) = 49$. Note that $\text{Cov}[X, Y] > 0$ is expected because X and Y are positively correlated.

Definition 5.7 **Correlation**

The *correlation* of X and Y is $r_{X,Y} = \frac{\text{Cov}[X,Y]}{\sqrt{\text{Var}[X]\text{Var}[Y]}}$

(a) $\text{Cov}[X, Y] = r_{X,Y} \sqrt{\text{Var}[X]\text{Var}[Y]}.$

(b) $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2 \text{Cov}[X, Y].$

(c) If $X = Y$, $\text{Cov}[X, Y] = \text{Var}[X] = \text{Var}[Y]$ and $r_{X,Y} = \frac{\text{Cov}[X,Y]}{\sqrt{\text{Var}[X]\text{Var}[Y]}} = \frac{\text{Var}[X]}{\sqrt{\text{Var}[X]\text{Var}[X]}} = \frac{\text{Var}[X]}{\text{Var}[X]} = 1.$

Example 5.17

For the integrated circuits tests in Example 5.3, we found in Example 5.5 that the probability model for X and Y is given by the following matrix.

$P_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$	0.01	0	0	0.01
$x = 1$	0.09	0.09	0	0.18
$x = 2$	0	0	0.81	0.81
$P_Y(y)$	0.10	0.09	0.81	

Find $r_{X,Y}$ and $\text{Cov}[X, Y]$.

$$\begin{aligned}r_{X,Y} &= \text{E}[XY] = \sum_{x=0}^2 \sum_{y=0}^2 xyP_{X,Y}(x,y) \\ &= (1)(1)0.09 + (2)(2)0.81 = 3.33.\end{aligned}$$

$$\text{E}[X] = (1)(0.18) + (2)(0.81) = 1.80,$$

$$\text{E}[Y] = (1)(0.09) + (2)(0.81) = 1.71.$$

$$\text{Cov}[X,Y] = 3.33 - (1.80)(1.71) = 0.252.$$

———— **Definition 5.8** ———— **Orthogonal Random Variables**

Random variables X and Y are *orthogonal* if $r_{X,Y} = 0$.

———— **Definition 5.9** ———— **Uncorrelated Random Variables**

Random variables X and Y are *uncorrelated* if $\text{Cov}[X, Y] = 0$.

